

LAB EXPERIMENT RECORD

Set 2: Customer Feedback Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

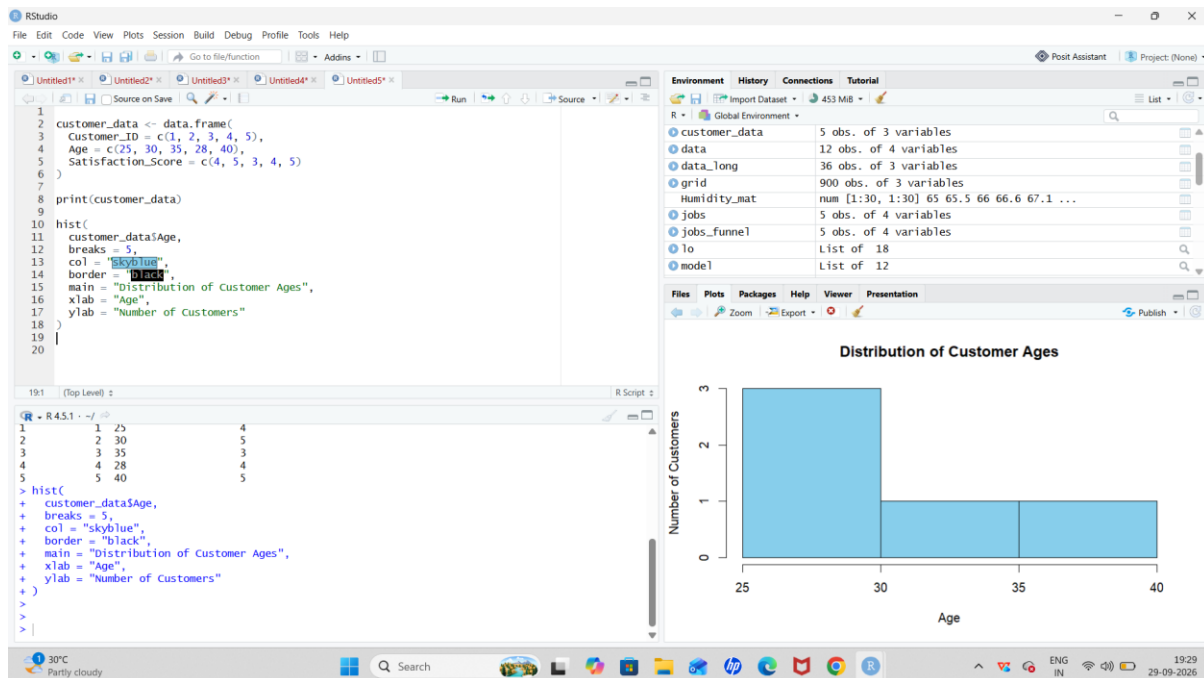
To analyze customer satisfaction data using R programming by creating a histogram, pie chart, stacked bar chart, and word cloud to understand customer age distribution, satisfaction levels, and common feedback sentiments.

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction_Score = c(4, 5, 3, 4, 5)  
)  
  
print(customer_data)  
  
hist(  
  customer_data$Age,  
  breaks = 5,  
  col = "skyblue",  
  border = "black",  
  main = "Distribution of Customer Ages",  
  xlab = "Age",  
  ylab = "Number of Customers"
```

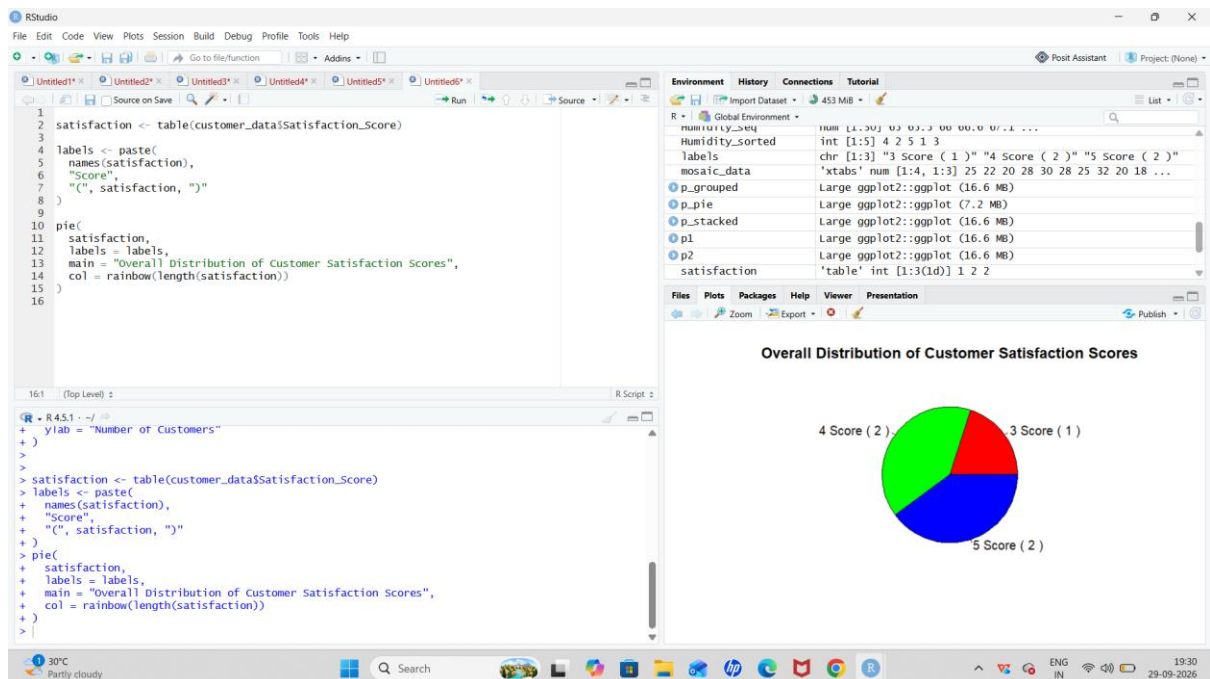


2. Generate a pie chart to display the overall distribution of customer satisfaction scores.

```
satisfaction <- table(customer_data$Satisfaction_Score)

labels <- paste(
  names(satisfaction),
  "Score",
  "(", satisfaction, ")")
)

pie(
  satisfaction,
  labels = labels,
  main = "Overall Distribution of Customer Satisfaction Scores",
  col = rainbow(length(satisfaction))
)
```

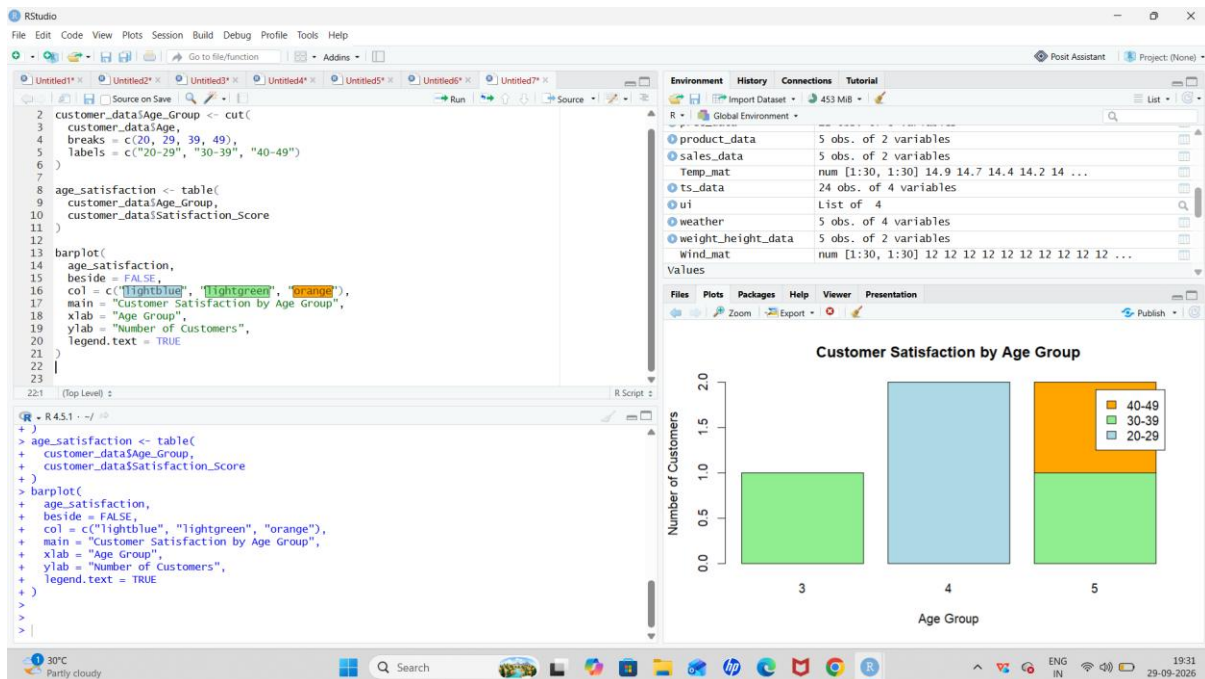


3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
customer_data$Age_Group <- cut(
  customer_data$Age,
  breaks = c(20, 29, 39, 49),
  labels = c("20-29", "30-39", "40-49")
)

age_satisfaction <- table(
  customer_data$Age_Group,
  customer_data$Satisfaction_Score
)

barplot(
  age_satisfaction,
  beside = FALSE,
  col = c("lightblue", "lightgreen", "orange"),
  main = "Customer Satisfaction by Age Group",
  xlab = "Age Group",
  ylab = "Number of Customers",
  legend.text = TRUE
)
```



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer

```
install.packages("tm")
install.packages("wordcloud")
install.packages("RColorBrewer")

library(tm)
library(wordcloud)
library(RColorBrewer)

feedback <- c(
  "Excellent service and very friendly staff",
  "I am very satisfied with the service",
  "Good customer support and quick response",
  "Excellent product quality",
  "Very happy with the service",
  "The service was good and satisfactory",
  "Friendly staff and excellent support",
  "I am satisfied with the product",
  "Good quality and excellent service",
  "Very happy with the customer service"
)

corpus <- Corpus(VectorSource(feedback))

corpus <- tm_map(corpus, content_transformer(tolower))

corpus <- tm_map(corpus, removePunctuation)

corpus <- tm_map(corpus, removeNumbers)

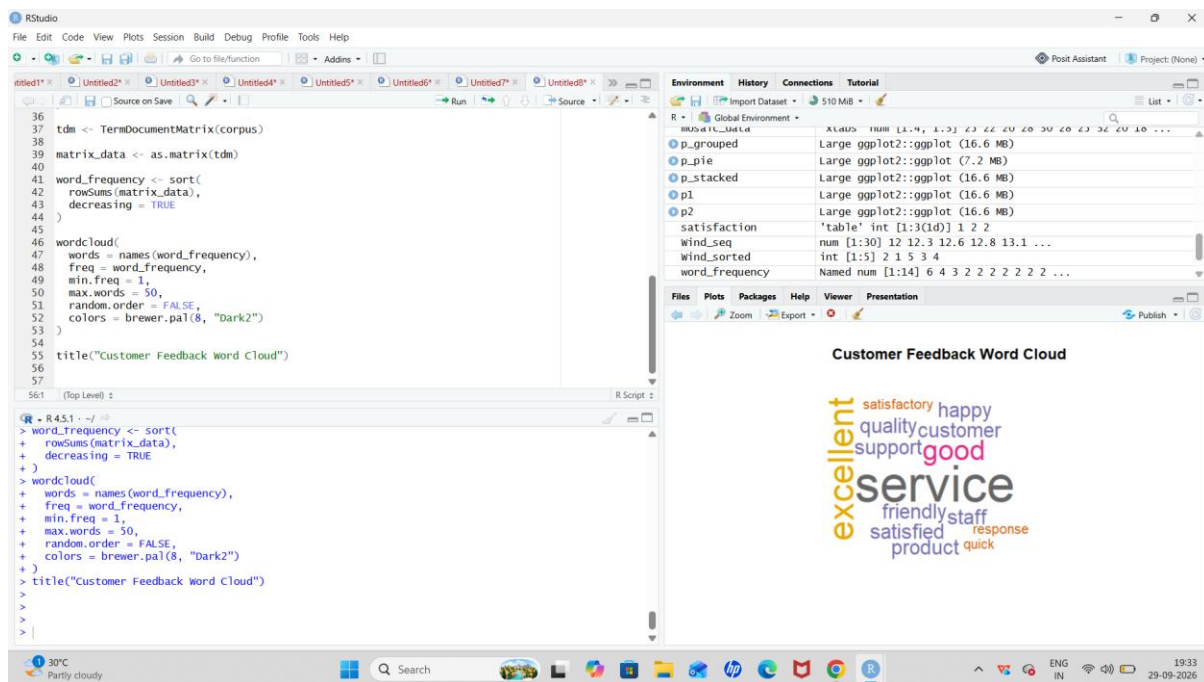
corpus <- tm_map(
  corpus,
  removeWords,
  stopwords("english")
)

tdm <- TermDocumentMatrix(corpus)

matrix_data <- as.matrix(tdm)

word_frequency <- sort(
  rowSums(matrix_data),
  decreasing = TRUE
)

wordcloud(
  words = names(word_frequency),
  freq = word_frequency,
  min.freq = 1,
  max.words = 50,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```



Procedure

1. Create the customer satisfaction dataset in R with Customer ID, Age, and Satisfaction Score.
2. Use the `hist()` function to create a histogram showing the distribution of customer ages.
3. Use the `pie()` function to create a pie chart representing the overall distribution of satisfaction scores.
4. Divide customers into different age groups using the `cut()` function.
5. Create a frequency table of age groups and satisfaction scores.
6. Use the `barplot()` function to generate a stacked bar chart showing satisfaction scores across different age groups.
7. Collect open-ended customer feedback and create a text corpus using the `tm` package.
8. Remove punctuation, numbers, and common stop words from the feedback.
9. Calculate word frequencies and use the `wordcloud` package to generate a word cloud.

10. Analyze the frequently occurring words to identify common customer sentiments.

Result

The customer feedback data was successfully analyzed using R. The **histogram** displayed the distribution of customer ages, while the **pie chart** showed the distribution of satisfaction scores. The **stacked bar chart** illustrated satisfaction levels across different age groups. The **word cloud** identified frequently occurring words in customer feedback, helping to understand the common customer sentiments and opinions.